Ateneo de Manila University

Archīum Ateneo

Ateneo School of Medicine and Public Health Publications

Ateneo School of Medicine and Public Health

1-22-2025

Impact of Primary Care Benefits on Healthcare Utilisation and Estimated Out-of-pocket Expenses in Urban, Rural and Remote Settings in the Philippines

Janelle Micaela S. Panganiban
University of the Philippines Diliman

Romelei Camiling-Alfonso
University of the Philippines Diliman

Josephine T. Sanchez
University of the Philippines Diliman

Regine Ynez H. De Mesa
University of the Philippines Diliman

Gillian Sandigan
University of the Philippines Diliman

Follow this and additional works at: https://archium.ateneo.edu/asmph-pubs
See next page for additional authors

Part of the Health Services Administration Commons

Custom Citation

Janelle Micaela S Panganiban, Romelei Camiling-Alfonso, Josephine T Sanchez, Regine Ynez H De Mesa, Gillian Sandigan, Arianna Maever L Amit, Mia P Rey, Johanna Faye E Lopez, Noleen Marie Fabian, Cara Lois T Galingana, Nannette Bernal-Sundiang, Maria Rhodora N Aquino, Chad Lester Lastrilla, Miguel Callo, Carol Stephanie Chua Tan-Lim, Leonila F Dans, Jose Rafael A Marfori, Ramon Pedro Paterno, Antonio L Dans - Impact of primary care benefits on healthcare utilisation and estimated out-of-pocket expenses in urban, rural and remote settings in the Philippines: BMJ Open Quality 2025;14:e002676.

This Article is brought to you for free and open access by the Ateneo School of Medicine and Public Health at Archīum Ateneo. It has been accepted for inclusion in Ateneo School of Medicine and Public Health Publications by an authorized administrator of Archīum Ateneo. For more information, please contact oadrcw.ls@ateneo.edu.

Authors

Janelle Micaela S. Panganiban, Romelei Camiling-Alfonso, Josephine T. Sanchez, Regine Ynez H. De Mesa, Gillian Sandigan, Arianna Maever L. Amit, Mia P. Rey, Johanna Faye E. Lopez, Noleen Marie Fabian, Cara Lois T. Galingana, Nannette Bernal-Sundiang, Maria Rhodora N. Aquino, Chad Lester Lastrilla, Miguel Callo, Carol Stephanie Chua Tan-Lim, Leonila F. Dans, Jose Rafael A. Marfori, Ramon Pedro Paterno, and Antonio L. Dans

This article is available at Archīum Ateneo: https://archium.ateneo.edu/asmph-pubs/298

BMJ Open Quality

Impact of primary care benefits on healthcare utilisation and estimated out-of-pocket expenses in urban, rural and remote settings in the Philippines

Janelle Micaela S Panganiban,[1] Romelei Camiling-Alfonso,[1] Josephine T Sanchez,[1] Regine Ynez H De Mesa ,[1] Gillian Sandigan,[1] Arianna Maever L Amit,[2,3] Mia P Rey,[1,4] Johanna Faye E Lopez,[1] Noleen Marie Fabian ,[1,5] Cara Lois T Galingana,[1] Nannette Bernal-Sundiang,[1] Maria Rhodora N Aquino,[1] Chad Lester Lastrilla,[1] Miguel Callo,[1] Carol Stephanie Chua Tan-Lim ,[1,6] Leonila F Dans,[1] Jose Rafael A Marfori,[1,6] Ramon Pedro Paterno,[1] Antonio L Dans[1,7]

To cite: Panganiban JMS, Camiling-Alfonso R, Sanchez JT, *et al*. Impact of primary care benefits on healthcare utilisation and estimated out-of-pocket expenses in urban, rural and remote settings in the Philippines. *BMJ Open Quality* 2025;**14**:e002676. doi:10.1136/bmjoq-2023-002676

► Additional supplemental material is published online only. To view, please visit the journal online (https://doi.org/10.1136/bmjoq-2023-002676).

Received 2 November 2023
Accepted 9 January 2025

Check for updates

© Author(s) (or their employer(s)) 2025. Re-use permitted under CC BY-NC. No commercial re-use. See rights and permissions. Published by BMJ Group.

For numbered affiliations see end of article.

Correspondence to
Noleen Marie Fabian;
nmcfabian@gmail.com

ABSTRACT

Background This study aimed to determine the effects of primary care interventions on healthcare utilisation and estimated out-of-pocket (OOP) expenses in selected urban, rural and remote settings in the Philippines.

Methodology Context-specific measures relating to expanding healthcare provider networks, augmenting the health human workforce and subsidising transportation costs were implemented to strengthen primary care systems. In this study, two key outcomes were monitored: (1) monthly healthcare utilisation measured by the total number of outpatient consultations per site and (2) change in OOP expenses from baseline to endline within a 1 year study period.

Results All sites had a positive trend in monthly outpatient consultations in healthcare utilisation over 1 year. The remote site had the steepest increase in outpatient consultations, with a 401% increase compared with the baseline during the peak of consultations at month 7. The urban site had a 62% increase in outpatient consultations from baseline to month 6, while the rural site had a 251% increase from baseline to month 11, which corresponded to the peak month in terms of the number of outpatient consultations. The rural site had the largest decrease in estimated OOP expenses (50.3% reduction, 95% CI −88 to −13), followed by the remote site (33.2% reduction, 95% CI −67,+1) and the urban site (16.0% reduction, 95% CI −65,+33).

Conclusion The rural site showed a significant reduction in estimated OOP expenses and an increase in healthcare utilisation. The remote site had the steepest increase in utilisation, but the reduction in estimated OOP expenses was not statistically significant. The urban site experienced the lowest increase in utilisation, and the smallest reduction in estimated OOP expenses, which was also not statistically significant. Implementing primary care benefits will necessitate contextualised approaches to avoid the inadvertent aggravation of inequities in healthcare.

WHAT IS ALREADY KNOWN ON THIS TOPIC

⇒ Primary care resources are finite. This study explores how resources can be used while preserving sustainability.

WHAT THIS STUDY ADDS

⇒ This study explores factors that may cause utilisation behaviours to vary across sites. It also explores how external actors change the utilisation behaviours of healthcare consumers.

HOW THIS STUDY MIGHT AFFECT RESEARCH, PRACTICE OR POLICY

⇒ This study enables policymakers to explore potential strategies to increase utilisation in varying localities.

INTRODUCTION

Inequitable access to health services remains a significant concern in the Philippines.[1] Social, economic and geographic barriers give rise to variations in access to quality healthcare services, resulting in meagre improvements in health outcomes across regions and economic strata.[2] Despite the implementation of policies to expand the coverage of health services, such as the National Health Insurance Act of 2013, the persistence of barriers to accessing quality healthcare services and high financial costs discourages healthcare utilisation particularly in resource-limited and GIDAs.[3–8]

In 2018, the government passed the Universal Healthcare (UHC) Law to address these gaps and strengthen primary care in the country.[9 10] To support the implementation of the UHC Law, the Philippine Primary Care Studies (PPCS) programme piloted primary care strengthening interventions in urban,

BMJ Open Quality: first published as 10.1136/bmjoq-2023-002676 on 22 January 2025. Downloaded from https://bmjopenquality.bmj.com on 7 April 2025 by guest. Protected by copyright, including for uses related to text and data mining, AI training, and similar technologies.

rural and remote settings in the Philippines to inform the government of strategies to realise and enable true universal health coverage for the country. The exposition of working models in selected urban, rural and remote sites aims to demonstrate to policymakers how to invest in primary care systems that are feasible, efficient, effective and equitable.

The primary care interventions are intended to address existing gaps in healthcare utilisation and health expenditure. Levesque *et al* explained that healthcare utilisation is a 'realised access' to care.[11] Additionally, Banaag *et al* state that reducing out-of-pocket (OOP) expenditure as a percentage of total health expenditure is pertinent in evaluating the efficiency of public financing schemes.[2] Therefore, these variables should become vital in evaluating the successful implementation of universal healthcare in the Philippines.

A previous study[12] conducted a survey to determine the baseline health benefit utilisation and OOP expenses in the three primary care settings of the PPCS prior to the implementation of the primary care strengthening interventions. This is a follow-up study to describe the observed changes in healthcare utilisation and estimated OOP expenses pre- and post-implementation of primary care interventions across selected settings in the Philippines. The specific objectives of the study are (1) to determine the effect of primary care interventions on healthcare utilisation across three study sites, an urban, rural and remote site and (2) to compare the change in estimated OOP expenses across the study sites pre- and post-intervention. It borrows Levesque *et al*'s Conceptual Framework on Access to Health, which elucidates supply-side and demand-side factors that impact healthcare utilisation.[11]

Levesque's conceptualisation of healthcare access considers the systemic variations between geographic locations, which closely aligns with the healthcare landscape in the Philippines. It highlights the study's interest in evaluating interventions to promote the equitable implementation of primary care services that consider the specific population's characteristics, resources and geographic location.[11] From a health equity perspective, structural barriers must be addressed so that interventions that intend to improve primary care systems can equitably impact communities across urban, rural and remote areas.[13–16] Assessing and comparing these experiences among urban, rural and remote patients will inform how context-specific barriers impact the full realisation of UHC and the promotion of health equity in the country.

METHODOLOGY

Study sites

The urban site was an academic institution in Metro Manila, which has a 25-bed outpatient facility that served 31 000 university-based students, faculty, employees and residents. The facility was affiliated with diagnostic and pharmacy providers within the university system, where most referrals were directly fulfilled. Outside of the facility, patients also had immediate access to other primary care providers within the metropolitan area.

The rural site was a fourth-class municipality (ie, municipalities with an average annual income of $60 000–1 00 000) with 35 298 residents across 14 barangays (administrative districts).

The remote site is a geographically isolated and disadvantaged area (GIDA), which, in the Philippines, is defined as a community with a marginalised population that is physically and socio-economically isolated from mainstream society.[17] The remote site had 23 932 residents across 24 barangays. Transportation and internet connectivity were challenges for this site.

For both rural and remote sites, there is a rural health unit (RHU) and barangay health stations (BHS) for its constituents. These public facilities had limited diagnostic capacity and pharmaceutical offerings. A physician supervised the RHU, while the BHS was staffed by barangay health workers (BHWs). Staffing for both sites was limited, with only one municipal health officer supported by midwives, nurses and BHWs before the study. Patients had limited access to other healthcare providers outside these public healthcare facilities.

Primary care interventions

PPCS introduced interventions to strengthen the primary care system in these three sites. These were implemented in the urban site from 2016 to 2017 and, subsequently, in the rural and remote areas from 2019 to 2020.

To address health financing problems, the study introduced a primary care benefit package worth PhP 350 per capita, with a Php 2000 spending cap per patient over 1 year. The amount could be used for any condition, that is, it was 'disease-agnostic'. It was also comprehensive because it could be used for consultations, diagnostic tests or prescribed medications. Patients were registered opportunistically at the point of care during their first consult. This contrasts with PhilHealth's existing process of prior registration, which is repetitive, resource-intensive and inefficient.[16]

An electronic health record (EHR) system was established to unify patient records and centralise medical information. Healthcare personnel used the system for triaging, referrals, medicine and laboratory request printing, medical certificate printing, drug dispensing, laboratory results releasing, financial tracking, patient census and record management.

Technical training on the most common primary care conditions was conducted for the healthcare workers, including healthcare professionals (doctors, nurses and midwives) and community health workers. They were also given free subscriptions and training on how to use a clinical decision support tool (UpToDate) to assist them in performing their duties.

Community engagement activities for patients were also conducted. Promotional materials such as brochures, tarpaulins and social media pages were cascaded to

BMJ Open Quality: first published as 10.1136/bmjoq-2023-002676 on 22 January 2025. Downloaded from https://bmjopenquality.bmj.com on 7 April 2025 by guest. Protected by copyright, including for uses related to text and data mining, AI training, and similar technologies.

Table 1 Summary of primary care interventions

Intervention	Urban	Rural	Remote
Health financing: implementation of a primary care benefit package that covered patient consults, medicines and laboratory services of up to Php 2000 per patient.	✓	✓	✓
Health information system: development of an electronic health record (EHR) system that centralised patient records and referral systems.	✓	✓	✓
Healthcare worker training: technical training on primary care and clinical practice guidelines and the provision of a clinical decision-making platform.	✓	✓	✓
Community engagement: dissemination of print and digital materials to raise public awareness of primary care, provider networks and risk sharing.	✓	✓	✓
Healthcare provider networks: expansion of provider availability through partnerships with private laboratories and pharmacies.		✓	✓
Health human resources: augmentation of health workforce by hiring additional physicians.		✓	✓
Transportation: subsidised transportation costs to and from the health facility.			✓

inform patients of the importance of primary care. Town hall meetings and information campaigns were also conducted to manage community expectations on primary care and prevent benefit misuse.

Context-specific interventions were implemented in rural and remote sites. Partnerships with private laboratories and pharmacies were established due to the limitations of public facilities in fulfilling the needed healthcare services. Additional healthcare providers were recruited due to the lack of physician staffing for both sites. Transportation costs were subsidised to address the geographical barriers in the remote area. These interventions are summarised in table 1.

Study design and overview

Descriptive trend analysis was used for healthcare utilisation. Healthcare utilisation was measured as the total number of outpatient consultations recorded in the EHR system at each site. These consultations included curative and preventive visits and were reported monthly.

Pre-post evaluation for OOP expenses. OOP expenses were measured using a healthcare benefit utilisation and financial risk protection questionnaire that was pre-tested and implemented in a prior study.[17] The follow-up period was October 2016 to September 2017 for the urban site, April 2019 to March 2020 for the rural site and May 2019 to April 2020 for the remote site. The start of the study period coincided with initiating and implementing the primary care interventions. Due to the urgency of disseminating the results on the impact of primary care interventions on these two critical outcomes to inform policy, we analysed and reported the results within a 1 year study period for each site.

Preparation of questionnaire for OOP expenses

A healthcare benefit utilisation and financial risk protection questionnaire was previously created to assess patients' previous consultations, admissions and expenses incurred within a prescribed recall period (online supplemental appendix 1).

For this study, we only extracted data on outpatient expenses. Participants were asked to estimate how much they spent on outpatient care, specifically consultations, diagnostic tests and medicines. They were also requested to identify their respective financing sources, such as OOP, national health insurance (provided through the study) or other sources. To minimise recall bias, a recall period of 3 months was set.

Sampling and survey distribution for OOP expenses

In the baseline study published in 2024, the sample size was calculated based on the two outcomes, namely, utilisation and OOP expenses. Since the sample size requirements were larger for the utilisation parameter, this was the sample size requirement used in the baseline study.[12] Simple random sampling was done for the urban site, while stratified sampling was done for the rural and remote sites where heads of the households were sampled from each barangay (stratum).

In this follow-up study, total enumeration was done for the utilisation outcome. The total number of outpatient consultations in each of the research sites was reported. For the OOP expense outcome, we invited the same respondents who answered the baseline survey to participate in the follow-up study. Respondents who refused to participate or could not be contacted were replaced with new respondents. The new respondents for replacement were matched by employment category and department in the urban site, and by sex, age and household site in the rural and remote sites.

All endline surveys were done at least 1 year after implementing the interventions. Written and verbal informed consent was collected from each respondent before survey administration.

Data analysis

Healthcare utilisation was measured by the number of outpatient consultations reflected on the EHR records across each site. The total number of monthly outpatient consultations was collected for each site and is presented

BMJ Open Quality: first published as 10.1136/bmjoq-2023-002676 on 22 January 2025. Downloaded from https://bmjopenquality.bmj.com on 7 April 2025 by guest. Protected by copyright, including for uses related to text and data mining, AI training, and similar technologies.

using a line graph. Meanwhile, estimated OOP expenses were based on questionnaire data. These were summarised in absolute values and percentage changes from baseline to endline periods. Data collected were encoded and processed using Microsoft Excel. The percentage change from baseline to endline was computed as

$$\%\ \text{change} = \frac{\text{average OOP expense per capita at endline} - \text{average OOP expense per capita at baseline}}{\text{average OOP expense per capita at endline}}$$

Conceptual framework of access to healthcare

In 2013, Levesque *et al* proposed a conceptual framework for understanding healthcare access.[11] The framework highlights five supply-side determinants namely: (1) approachability, which refers to the ability of individuals with health needs to recognise the existence of available services, access them and benefit from their impact on their health; (2) acceptability, which involves cultural and social factors that influence people's willingness to accept certain aspects of a service, such as the provider's sex or social group, and the beliefs related to medical systems; (3) Availability and Accommodation, which mean that health services, including the physical facilities and healthcare professionals, can be accessed both physically and promptly; (4) Affordability, which indicates people's financial ability to allocate resources and time to access appropriate services; and (5) Appropriateness, which refers to how well services meet clients' needs, including their timeliness, the thoroughness of health problem assessments, the accuracy of treatments and the technical and interpersonal quality of the services provided.

Meanwhile, five demand-side determinants also shape healthcare utilisation and access, which complements the five supply-side determinants, respectively. Levesque identifies the following: (1) the ability to perceive, influenced by factors such as health literacy, knowledge about health and beliefs related to health and illness; (2) the ability to seek, which pertains to personal autonomy and the capacity to choose to pursue care; (3) the ability to reach, which involves personal mobility, access to transportation, occupational flexibility and awareness of health services, all of which enable an individual to physically access service providers; (4) the ability to pay, which refers to the capacity to generate economic resources without incurring catastrophic expenses that would jeopardise the ability to meet basic necessities; and 5) the ability to engage, which involves client's participation in decision-making and treatment choices.

For this study, Levesque's framework was used to examine the influence of study interventions on healthcare utilisation. Supply-side interventions of the study respond to identified site-specific gaps among the determinants of healthcare utilisation (figure 1). Self-reported OOP expenses are considered a proxy for patients' ability to pay (demand side) in response to the primary care benefit package introduced by the study (supply side).

RESULTS

Healthcare utilisation

Figure 2 summarises the monthly outpatient consultations in each of the research sites. The urban site had a positive trend in outpatient consultations throughout the 1 year study period. There were 510 consultations recorded at baseline (month 1). At endline (month 12), there were 663 consultations, with a total of 7664 consults recorded within the 1 year study period. Monthly variations in the number of outpatient consultations can be observed. A 62% increase in utilisation was observed between baseline and March 2017 (month 6), when the number of outpatient consultations peaked.

The rural site also had a positive trend in outpatient consultations in the first year of the implementation of

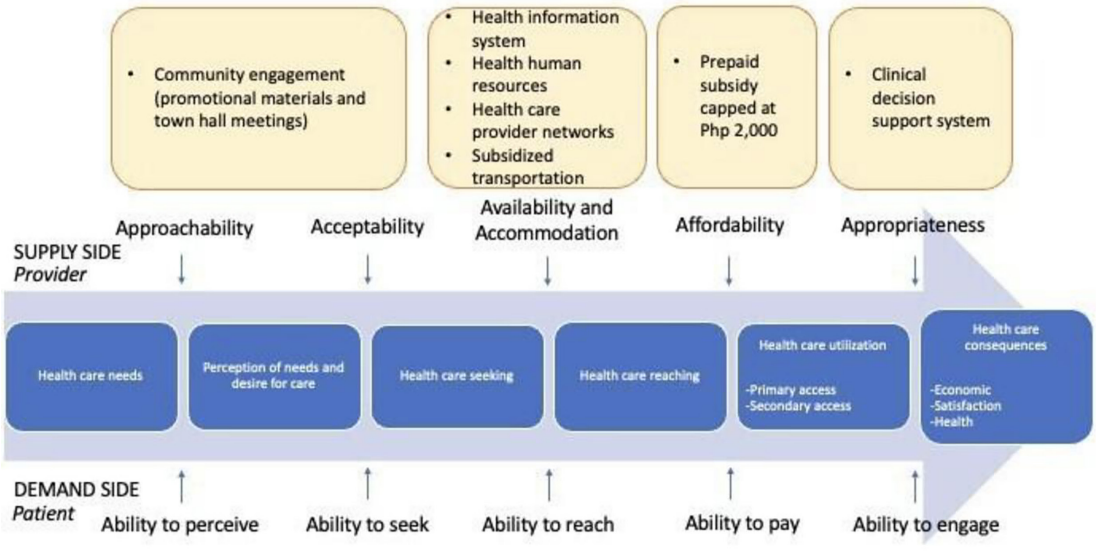


Figure 1 Levesque's conceptual framework of access to healthcare.[11]

BMJ Open Quality: first published as 10.1136/bmjoq-2023-002676 on 22 January 2025. Downloaded from https://bmjopenquality.bmj.com on 7 April 2025 by guest. Protected by copyright, including for uses related to text and data mining, AI training, and similar technologies.

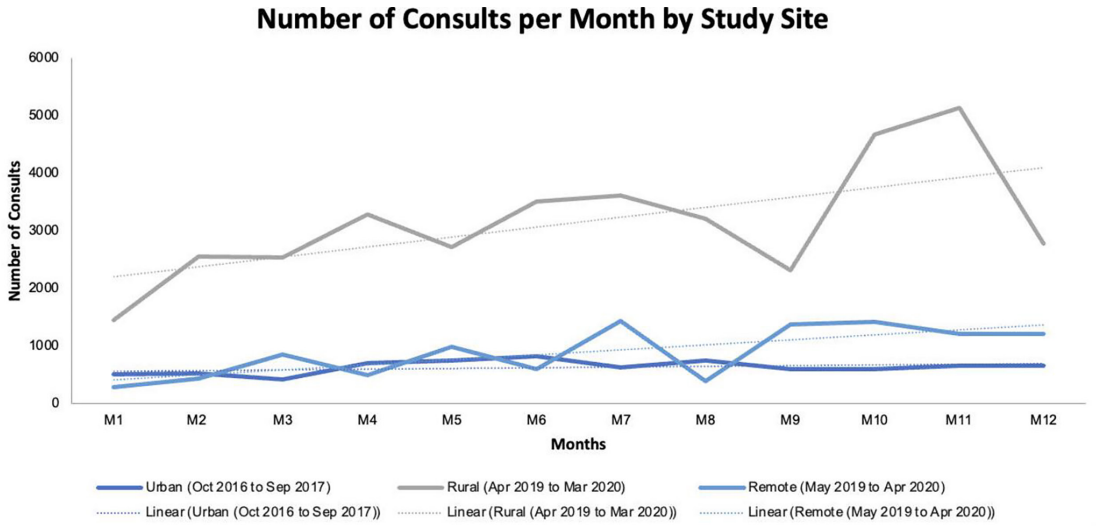


Figure 2 Monthly outpatient consultations across research sites (year 1). The solid lines represent the number of outpatient consults per site. The dotted lines represent the slopes of the lines per site.

primary care interventions. There were 1460 consultations recorded at baseline (month 1). At endline (month 12), there were 2779 consultations. A total of 37818 consultations were recorded in the 1 year study period for the rural site, which is the highest among the three sites. The number of consultations peaked in February 2020 (month 11), with a 251% increase in outpatient consultations compared with the baseline. Like the urban site, there are monthly variations in outpatient consultations.

The remote site had a positive trend in the number of monthly outpatient consultations, with monthly variations noted as well. There were 287 consultations recorded at baseline (month 1). At endline (month 12), there were 1209 consultations. A total of 10727 consultations were recorded in the 1 year study period for the remote site. The number of consultations peaked in November 2019 (month 7) at 1439, a 401% increase from baseline.

Pre- and post-intervention evaluation of estimated OOP expenses

The number of respondents at baseline and endline is summarised in table 2. The urban sample comprised faculty members, employees and contractual workers, while the rural and remote samples included household heads in the catchment populations. At endline, we only had 202 urban respondents despite attempts to replace respondents.

Table 2 Number of study participants at baseline and endline

	Baseline	Endline	Sample composition
Urban	358	202	University-based faculty, employees, contractual workers
Rural	228	228	Household heads
Remote	202	202	Household heads
Total	788	632	

Table 3 summarises the change in estimated OOP expenses from baseline to endline periods across the study sites. Compared with baseline, the rural site had the largest decrease in estimated OOP expenses, with a 50.3% reduction. The remote site had a 33.2% decrease, while the urban site had the smallest decrease (16.0%) in estimated OOP expenses compared with baseline. Only the rural site had a statistically significant reduction in estimated OOP expenses at endline.

DISCUSSION

The rural site had the highest number of cumulative outpatient consultations (37818 consultations), the largest decrease in average estimated outpatient OOP expense from baseline to endline (−50.3%). The urban site had the smallest increase in outpatient consultations and smallest reduction in estimated outpatient OOP expense from baseline to endline.

Urban healthcare utilisation and estimated OOP expenses

The urban site had minimal increases in monthly outpatient consultations throughout the study period. Two things might explain this lack of response. First, introducing relatively small subsidies for primary care may not significantly alter healthcare utilisation in a community with regular access. Second, the baseline utilisation may already represent the optimised healthcare use for the site.

The urban site has an open and pluralistic system, highlighting the multiplicity and availability of stakeholders engaged in healthcare services.[17] Multiple health service providers exist outside the urban site, where patients may opt to avail of their health services. This can explain the minimal increase in the number of outpatient consultations for the site. Moreover, the urban site, relative to its rural and remote counterparts, may already have better health outcomes due to better access to health services.

BMJ Open Quality: first published as 10.1136/bmjoq-2023-002676 on 22 January 2025. Downloaded from https://bmjopenquality.bmj.com on 7 April 2025 by guest. Protected by copyright, including for uses related to text and data mining, AI training, and similar technologies.

Table 3 Change in estimated out-of-pocket expenses (PhP) from baseline to endline (Y1)

	Average estimated outpatient out-of-pocket expense per capita per year		
	Baseline in Php (95% CI)*	Endline in Php (95% CI)	Change in % (95% CI)
Urban (N=202)	521 (379, 663)	438 (225, 650)	–16.0 (-65,+33)
Rural (N=228)	308 (207, 409)	153 (99, 207)	–50.3 (-88, -13)
Remote (N=202)	555 (406, 705)	371 (254, 483)	–33.2 (-67,+1)

*The baseline values slightly differ from the previously published study. Values were recalculated based on the actual average household size in the rural and remote sites and the PhilHealth data on average household size for the urban site. The previously published study used rough estimates of household size.[12]

Vlahov *et al* explore the urban health 'advantage' phenomenon and argue that health indicators are better in urban areas due to socioeconomic heterogeneity, higher levels of social support and higher access to necessities of life.[18] This likely reduces the need for patients to visit the urban site, compared with rural and remote sites. Grounded on the health access framework by Levesque *et al*, we see that the urban healthcare system, with present and substantial resources under the Availability and Accommodation domain, led to stability in healthcare utilisation for the urban site.

Varied results were noted for the average estimated OOP expense of patients in the urban site. While the average estimated OOP expenses reduced by 16.0% from baseline to year 1, the percentage of estimated OOP expense at the end of year 1 remains high at 53.7%. We can attribute the reduction in estimated OOP expenses to the Php 2000 benefit package provided by the programme to increase patients' ability to pay.[11] However, it is still likely that some patients did not maximise their benefits at the urban site clinic and instead acquired care from other service providers not covered by the benefit package. Similarly, this can be explained by the open and pluralistic system of the urban site, where patients are not obliged to seek care from one provider alone.

Rural healthcare utilisation and estimated OOP expenses

The rural site had a steep increase in monthly outpatient consultations, demonstrating higher utilisation. This site also had a large reduction in estimated OOP expenses. Considering Levesque *et al*'s health access framework, we may relate this to the supply-side interventions of the study targeting the areas of Approachability, Availability and Accommodation, and Affordability.

Approachability towards the healthcare site was improved using promotional activities, such as the cascade of flyers and town hall meetings. The structure of these activities included presenting the basic concepts surrounding primary care and emphasising its importance in improving patient health. These methods likely encouraged patients to visit the clinic more. Gupta *et al* describe the importance of health education materials, mainly through social marketing, to improve patient behaviour in seeking primary care.[19] Moreover, additional RHU physicians were recruited in partnership with the local government unit. The support of the local government was crucial in ensuring that the supply of healthcare providers was sufficient to meet healthcare demands.[16] An EHR system was also implemented, optimising the time spent by healthcare staff serving patients instead of managing patient records.

Improved average estimated OOP expense was also observed in the rural site between baseline and year 1, where the average estimated OOP expense was reduced by 50.3%. The percentage of estimated OOP expense at the rural site was the lowest across all sites at 18.3%. Like the urban site, these improvements can be attributed to providing a Php 2000 benefit package. Patients likely maximised this benefit, thus decreasing their OOP expenses. Furthermore, the study expanded the healthcare provider network of the rural site by partnering with laboratory and diagnostic providers. With additional providers in the system, patients' needs from these facilities were likely covered by the benefit package, thus decreasing their OOP expenses. These interventions can explain the improvement in OOP expense for health services.[11]

Remote healthcare utilisation and estimated OOP expenses

The remote site had fluctuating utilisation, as shown in their monthly outpatient consultations. The varying levels of health utilisation can be related to similar study interventions identified at the rural site. These initiatives targeted the areas of Approachability, Availability and Accommodation, and Affordability. However, the impact on the remote site is not as large as observed on the rural site. The existing structural barriers can largely explain this difference for the remote site.

The EHR system was not fully optimised due to poor internet connection. An effective EHR system is expected to support service delivery networks by (1) centralising patient information, (2) making comprehensive data available at every point of patient care and (3) ensuring accessibility from multiple locations and units.[20] Effective EHR implementation necessitates sufficient infrastructure.[21] The rural site had a good internet connection since the provincial government installed additional towers with long-distance routers to integrate the entire municipality under one network.[16] However, this was not feasible in the remote site due to its dense and mountainous landscape.[22] As a result, the remote site experienced considerable delays in addressing internet connectivity issues,

BMJ Open Quality: first published as 10.1136/bmjoq-2023-002676 on 22 January 2025. Downloaded from https://bmjopenquality.bmj.com on 7 April 2025 by guest. Protected by copyright, including for uses related to text and data mining, AI training, and similar technologies.

limiting access to operational community health stations and primary care services. Subsidised transportation costs to and from the health facility became crucial in ensuring primary care accessibility at the remote site. In contrast, the rural site could activate its community health stations through EHR implementation, embodying the essence of primary care of being as close to the community as possible.[23]

A 33.2% decrease in the average estimated OOP expense was noted between baseline and endline. Similar to the rural site, these improvements can be attributed to the provision of the Php 2000 benefit package that decreased the OOP expenses of patients. The healthcare provider network of the remote site was also expanded through partnerships with laboratory and diagnostic providers, allowing patients' needs from these facilities to be covered by the benefit package.

However, OOP expenses remained high at endline similar to the urban site. It is important to contextualise that healthcare spending in the remote site was low at baseline compared with the rural and urban sites. Therefore, health services were likely underused at the beginning, and patients were more likely to have more severe or complicated conditions. Chen *et al* highlighted geographical disparities in the utilisation of preventive care services, which can explain this underutilisation.[24] When patients were encouraged to consult more, health expenditures from outpatient consultations, laboratory services and medicines likely increased. The expansion of healthcare provider networks could have been the cost driver, such that some expenses from laboratories and pharmacies probably exceeded patient benefits.

Implications for UHC implementation

Equitable implementation of UHC in the Philippines should consider the variations in health system readiness across site-specific contexts. The contexts of each site are different, where the most disadvantaged remote site was already predisposed to equity concerns, specifically poor infrastructure.

The existing infrastructure impacts primary care utilisation across sites. Infrastructure elements, such as internet connectivity and geographical characteristics, affect the proliferation of primary care access points. These issues are most pressing in the remote site, where primary care options remained limited at baseline and endline because of the inefficient implementation of the EHR due to poor internet connectivity. Meanwhile, the rural site resolved this and activated more access points through community health stations. On the other hand, the urban site already had multiple primary care access points outside the study setting where patients could have otherwise received care.

Returning to Levesque *et al*'s framework for healthcare access,[11] existing infrastructure problems at the remote site limited the impact of the study's interventions under the Availability and Accommodation domain. With this domain not fully implemented, it is also likely that the effect of the Affordability domain, which follows it, remains unmet. Each domain of the framework must be addressed to achieve our vision for optimised healthcare utilisation and improved healthcare access. This calls for additional funding to ensure health system readiness across different geographic regions to adopt primary care and universal health coverage.

Limitations

The scope of this study is limited to outpatient consultations in RHUs for rural and remote sites and a primary care clinic in an urban site. Future studies may include data from community health stations, diagnostic laboratories and pharmacies. Another limitation is that not all consultation data points can be considered independent, as several individuals may likely have consulted several times during the study period. Clustering within households in the rural and remote may also have affected the number of consultations, since the observation units for this variable are individuals, while the sampling units are households. Moreover, the OOP expenses are estimated by patients through survey data. Recall bias might occur for some of these metrics. The potential impact of socio-demographic factors (eg, ethnicity, income level, family size) on the resulting data set remains outside the study's scope. Future studies may want to explore external factors that may ultimately contribute to patients' care utilisation. Lastly, the COVID-19 pandemic was declared in March 2020, which likely affected healthcare utilisation and OOP expenses towards the tail end of the first year of primary care implementation for the rural and remote sites.

CONCLUSION

This study measured healthcare utilisation and estimated OOP expenses across urban, rural and remote settings in the Philippines after introducing primary care interventions. The rural site showed the most promising results at the end of the study period for healthcare utilisation and estimated OOP expenses. In contrast, the urban site had a minimal increase in utilisation. This may be due to an open and pluralistic healthcare system, which provided several other options for healthcare. Due to existing infrastructure problems, the remote site could not fully optimise the interventions provided despite the provision of equal funds for primary care services. Such pre-existing problems in access may require the allocation of greater funds for disadvantaged areas and may necessitate contextualised approaches to avoid the inadvertent aggravation of inequities in healthcare access and utilisation.

Author affiliations
[1]Program on Health Systems Development - Philippine Primary Care Studies, Center for Integrative and Development Studies, University of the Philippines Diliman, Quezon City, Philippines

BMJ Open Quality: first published as 10.1136/bmjoq-2023-002676 on 22 January 2025. Downloaded from https://bmjopenquality.bmj.com on 7 April 2025 by guest. Protected by copyright, including for uses related to text and data mining, AI training, and similar technologies.

[2]National Clinical Trials and Translation Center, National Institutes of Health, University of the Philippines Manila, Manila, Philippines
[3]Center for Research and Innovation, School of Medicine and Public Health, Ateneo de Manila University, Pasig City, Philippines
[4]Department of Accounting and Finance, Cesar E A Virata School of Business, University of the Philippines Diliman, Quezon City, Philippines
[5]University of the East Ramon Magsaysay Memorial Medical Center Inc, Quezon City, Philippines
[6]Department of Clinical Epidemiology, College of Medicine, University of the Philippines Manila, Manila, Philippines
[7]College of Medicine, University of the Philippines Manila, Manila, Philippines

Acknowledgements The authors would like to express their gratitude to the staff of the University Health Service of the University of the Philippines Diliman, the local government unit of the Municipality of Samal, Province of Bataan and the local government unit of the Municipality of Bulusan, Province of Sorsogon, in supporting the success of this study.

Contributors JMSP and RCA spearheaded the conceptualisation and drafting of the manuscript. JTS, RYHDM, GS, AMLA, MPR, JFEL, NMF, CSCT-L and CLTG contributed in drafting the manuscript. NB-S, MRNA, CLL, MC, LFD, JRAM, RPP and ALD provided their insights during the data gathering and analysis stage. ALD will also serve as the guarantor of this manuscript.

Funding The study was supported by The Philippine Department of Health (DOH), the Philippine Health Insurance Corporation (PhilHealth), the Emerging Interdisciplinary Research Program (EIDR), the Center for Integrative and Development Studies (CIDS), and the Philippine Council for Health Research and Development (PCHRD).

Competing interests None declared.

Patient and public involvement Patients and/or the public were not involved in the design, conduct, reporting or dissemination plans of this research.

Patient consent for publication Not applicable.

Ethics approval This study involves human participants and was approved by the University of the Philippines Manila Research Ethics Board (UPMREB); Research No. 20-15-489-01. Participants gave informed consent to participate in the study before taking part.

Provenance and peer review Not commissioned; externally peer reviewed.

Data availability statement Data are available upon reasonable request.

Supplemental material This content has been supplied by the author(s). It has not been vetted by BMJ Publishing Group Limited (BMJ) and may not have been peer-reviewed. Any opinions or recommendations discussed are solely those of the author(s) and are not endorsed by BMJ. BMJ disclaims all liability and responsibility arising from any reliance placed on the content. Where the content includes any translated material, BMJ does not warrant the accuracy and reliability of the translations (including but not limited to local regulations, clinical guidelines, terminology, drug names and drug dosages), and is not responsible for any error and/or omissions arising from translation and adaptation or otherwise.

Open access This is an open access article distributed in accordance with the Creative Commons Attribution Non Commercial (CC BY-NC 4.0) license, which permits others to distribute, remix, adapt, build upon this work non-commercially, and license their derivative works on different terms, provided the original work is properly cited, appropriate credit is given, any changes made indicated, and the use is non-commercial. See: http://creativecommons.org/licenses/by-nc/4.0/.

ORCID iDs
Regine Ynez H De Mesa http://orcid.org/0000-0001-8011-8567
Noleen Marie Fabian http://orcid.org/0000-0002-8028-6242
Carol Stephanie Chua Tan-Lim http://orcid.org/0000-0001-8815-4191

REFERENCES

1 Dayrit MM, Lagrada LP, Picazo OF, *et al*. The Philippines health system review. *Health Syst Transit* 2018;8. Available: https://apps.who.int/iris/handle/10665/274579
2 Banaag MS, Dayrit MM, Mendoza RU. Health inequity in the Philippines. In: Batabyal A, Higano Y, Nijkamp P, eds. *Disease, Human Health, and Regional Growth and Development in Asia. New Frontiers in Regional Science: Asian Perspectives*. Singapore: Springer, 2019: 163–74.
3 Alipio M. A path analysis examining the relationship between access barriers to health services and healthcare utilization among the publicly insured: insights from a multiprovince survey in the philippines. *Public Health* d6vbm [Preprint] 2020.
4 Hodge A, Firth S, Bermejo R 3rd, *et al*. Utilisation of health services and the poor: deconstructing wealth-based differences in facility-based delivery in the Philippines. *BMC Public Health* 2016;16:523.
5 Sigua JA, Ong MM, Nuevo CE, *et al*. An introduction to the Philippine universal health care law, 2020. Available: https://thinkwell.global/wp-content/uploads/2020/05/PH-UHC-Law-Series_Brief-1.pdf [Accessed 23 Nov 2022].
6 Philippine Statistics Authority, Republic of the Philippines. Philippine national health accounts. n.d. Available: https://psa.gov.ph/statistics/pnha
7 Lasco G, Yu VG, David CC. The Lived Realities of Health Financing: A Qualitative Exploration of Catastrophic Health Expenditure in the Philippines. *Acta Med Philipp* 2022;56. Available: https://actamedicaphilippina.upm.edu.ph/index.php/acta/article/view/2389
8 Ulep VGT, Uy J, Casas LDD. Primary health care for noncommunicable diseases in the Philippines. Philippine Institute for Developmental Studies; 2020. Available: https://pidswebs.pids.gov.ph/CDN/PUBLICATIONS/pidsdps2039.pdf [Accessed 28 Feb 2024].
9 Congress of the Philippines. Republic act no. 11223 - an act instituting universal health care for all filipinos, prescribing reforms in the health care system, and appropriating funds therefor. 2018. Available: https://www.officialgazette.gov.ph/2019/02/20/republic-act-no-11223/ [Accessed 23 Nov 2022].
10 Department of Health Philippines. UHC. n.d. Available: https://doh.gov.ph/uhc/
11 Levesque JF, Harris MF, Russell G. Patient-centred access to health care: conceptualising access at the interface of health systems and populations. *Int J Equity Health* 2013;12:18.
12 Dans LF, Marfori JRA, De Mesa RYH, *et al*. Health Benefit Utilization and Out-of-Pocket Expenses in Outpatient Care and Hospitalizations: Baseline Surveys of Three Primary Care Sites in the Philippines. *Acta Med Philipp* 2024;58:133–40.
13 Dutton D. Financial, organizational and professional factors affecting health care utilization. *Soc Sci Med* 1986;23:721–35.
14 Jacobs B, Ir P, Bigdeli M, *et al*. Addressing access barriers to health services: an analytical framework for selecting appropriate interventions in low-income Asian countries. *Health Policy Plan* 2012;27:288–300.
15 Naria-Maritana MJN, Borlongan GR, Zarsuelo M-AM, *et al*. Addressing Primary Care Inequities in Underserved Areas of the Philippines: A Review. *Acta Med Philipp* 2020;54.
16 Bernal-Sundiang N, Mesa RYH, Marfori JRA, *et al*. Governance in Primary Care Systems: Experiences and Lessons from Urban, Rural, and Remote Settings in the Philippines. *Acta Med Philipp* 2023;57.
17 Han GS. The Myth of Medical Pluralism: A Critical Realist Perspective. *Sociol Res Online* 2002;6:65–80.
18 Vlahov D, Galea S, Freudenberg N. The urban health “advantage”. *J Urban Health* 2005;82:1–4.
19 Gupta A, Naidu NK, Kakkar R. Role of social marketing in promoting primary care to succeed in current era. *J Family Med Prim Care* 2019;8:3086.
20 Alipour J, Erfannia L, Karimi A, *et al*. Electronic Health Record Acceptance: A Descriptive Study in Zahedan, Southeast Iran. *J Health Med Inform* 2013;04:2–5.
21 Haux R. Health information systems - past, present, future. *Int J Med Inform* 2006;75:268–81.
22 Binaday J, Amarga A, Barrameda E, *et al*. Amphibians and reptiles in the vicinity of Bulusan lake, Bulusan Volcano natural park, Sorsogon, Philippines. *Philipp J Sci* 2017;146:339–51.
23 Safran DG. Defining the future of primary care: what can we learn from patients? *Ann Intern Med* 2003;138:248–55.
24 Chen L, Lu Y, Cao Q. The demand for preventive care services of residents in China and its influencing factors. *Chinese General Practice* 2015;18:428–32.

BMJ Open Quality: first published as 10.1136/bmjoq-2023-002676 on 22 January 2025. Downloaded from https://bmjopenquality.bmj.com on 7 April 2025 by guest. Protected by copyright, including for uses related to text and data mining, AI training, and similar technologies.